

Important parameters to be considered while making the selection

1. Consider the operating environment to evaluate if the processor is susceptible to external radiation, thermal management aspects, operating temperature range and more.
2. Performance requirements for the processing power—operating frequency, processing efficiency DMIPS/MHz, processing capability in terms of MIPS or DMIPS and support for multi-core and application-oriented co-processors/accelerators.
3. Availability of development tools and platforms (reference design).
4. Form factor expectations that include end-product physical size and physical weight.
5. Technical and long-term support available for the processor. This includes support for software infrastructure (OS, RTOS, firmware development, etc.).
6. Options available in the processor family for expansion in the long run. This could be for scaling up or down the cost and features with the same footprint.
7. Availability and cost of technologies for manufacturing the boards based on the selected processor.

Courtesy: Vijay Bharat S. and Sachidananda Karanth, lead architects, Mistral Solutions

interrupt latency, which can affect the real-time schedulability of the system,” explains Praveen.

Overall, one must keep in mind the processing bandwidth, memory bandwidth and memory sub-system interfaces. Depending on the application, you might require high-speed interfaces or processors, or even chip memory, etc.

Nilesh Ranpura, project manager, eInfoChips, shares some tips for us. “The first thing to be considered is the interface protocol meeting timings. This means, memory interfacing with processor needs to completely read and write cycles in efficient cycles. Write mem_read/write routine at low-level firmware which is tested on actual hardware.”

Shanmugasundaram adds that ultimately it all depends upon the application. “If an application is a real-time one, then bandwidth requirements are important along with prevention of system failures and data corruption. If the application is a non-real-time one, then the bandwidth requirements can take a backseat to avoid system failures and data corruption.”

Selecting a processor for embedded designs

Advent of different processor architectures has led to development of many application-specific processors. This section gives you an insight on selection of the correct processor based on the application for which it is required.

Another development worth noting in the Indian chip design scene is the

indigenous SoC designing being done by firms such as Sankhya Technologies and Dreamchip to develop a ready-to-use synthetic processor and SoC platform for major markets. “These allow the designers to make optimal hardware and software trade-offs that gain them the flexibility of software and the performance of hardware, which any embedded system design house would love to have,” explains Gopi Kumar Bulusu, chief technologist at Sankhya Technologies.

Table in the next page provides a comparison of the most popular processors that are being used by OEMs for various embedded applications.

PC-compatible CoMs

Computer on module, or COM, is a compact, high-performance system designed to focus on core competencies. Highly integrated PC-compatible COMs are based on x86 CPU architectures that provide a scalable solution and at the same time meet the advanced CPU application development needs. Some of the widely used x86 architecture processors are listed in the table. From Intel Atom series via AMD designs to Intel core processors, designers have a wide range of processors to select from, for designing their COMs.

While single-core processors from Intel Atom series like N270, N455 and Z510 are more focussed on balance in power consumption, performance and pricing, newer AMD G-series processors provide new graphic performance on us-

er-friendly interfaces or HD video onto the designs in single as well as dual-core options. AMD G-series with its 40nm technology vs Intel's 45nm technology provides a powerful yet cooler design. If you are going for a design with low-power consumption, Intel Atom Z510/N270 is the option to be locked, while if you need higher graphic performance in your designs, AMD G-series will fit best.

Low-power COMs

COMs that are specifically designed for high performance in low-power envelope are based on ARM processors, but other low-power x86 architectures could also be used. ‘SMARC’ or ‘Smart Mobility ARChitecture’ is a specification by the Standardisation Group of Embedded Technologies for such COMs.

ARM architecture-based processors are built with fewer transistors than other traditional x86 processors, resulting in reduced cost, heat and power usage. This makes them a perfect fit for the low-power COM designs. ARM Cortex A9-based i.Mx 6 family from Freescale is an example that supports a wide range of applications with its single, dual and quad core offerings.

Fanless box PC

The design of fanless box PCs used for various machine-controlled industrial applications is focussed on harsh environmental conditions and factory usage; however, they can also be designed for graphic-oriented applications like digital signage and media playback.

AMD G-series is a good option for graphic-oriented designs. Intel Atom series processors are good for robust low-power applications with a wide range of processor speeds to choose from. Apart from Intel Atom series and AMD G-series, VIA Nano X2 E and Eden E series are other dual-core series of processors that can be taken into account while designing fanless systems. X86 architecture-based Nano and Eden series have 40nm manufacturing technology and can withstand a wide range of temperatures due to their design.